THE UPPER SHRIEK OF A PERFECT HOMOMORPHISM IS TENSORING WITH ITS VALUE ON THE STRUCTURE SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-dualizing/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The perfectness of the target as a module over the chosen polynomial ring in the perfect case. Invoked inside the counted proof (source lines 4901-4903). Supplies the hypothesis that makes the tensor description of the internal hom available.
- The description of the internal hom out of a perfect module as tensoring with the internal hom of the module. Invoked inside the counted proof (source lines 4905-4906). Rewrites the internal hom defining the shriek as a tensor product.
- The double base change identity for the derived tensor product along the two presentations. Invoked inside the counted proof (source lines 4905-4907). Reassembles the tensor factors into tensoring with the shriek of the structure sheaf.

Source reference. Original lemma lemma-RHom-is-tensor-special: sources/stacks-dualizing/pinned-original-source.tex, source lines 2582–2582. Navigation pointer only: the counted text refers to the Stacks item lemma-RHom-is-tensor-special, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference more-algebra-lemma-double-base-change: sources/stacks-dualizing/pinned-original-source.tex, source lines 4695–4695. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-double-base-change, which belongs to another chapter of the project and is not part of this

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Source reference. Original reference more-algebra-lemma-flat-finite-presentation-perfect: sources/stacks-dualizing/pinned-original-source.tex, source lines 4771–4771. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-flat-finite-presentation-perfect, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference more-algebra-lemma-perfect-ring-map: sources/stacks-dualizing/pinned-original-source.tex, source lines 4650–4650. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-perfect-ring-map, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $\varphi : R \to A$ be a perfect homomorphism of Noetherian rings (for example φ is flat of finite type). Then $\varphi^!(K) = K \otimes^{\mathbf{L}}_R \varphi^!(R)$ for $K \in D(R)$.*

Proof. (The parenthetical statement follows from More on Algebra, Lemma more-algebra-lemma-flat-finite-presentation-perfect.) We can choose a factorization $R \to P \to A$ where P is a polynomial ring in n variables over R and then A is a perfect P-module, see More on Algebra, Lemma more-algebra-lemma-perfect-ring-map. Recall that $\varphi^!(K) = R\mathrm{Hom}(A, K \otimes^{\mathbf{L}}_R P[n])$. Thus the result follows from Lemma lemma-RHom-is-tensor-special and More on Algebra, Lemma more-algebra-lemma-double-base-change. □